

Sphere n -Point NS Blocks in the Pillow Frame and the Proposed Elliptic h -Recursion

Machine note

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Abstract

We give a self-contained recipe for bottom-component Neveu–Schwarz sphere blocks in the comb, or open-necklace, channel. The ordinary pillow map and its geometric conformal factor are retained. After primary propagation powers are removed, the proposed regular part is a universal infinite product; the singular part is determined by NS null-vector factorization and closes the elliptic recursion. We use the conventions of `Human Notes/SCblock.tex`, including its normalized elliptic integral, torus coordinate, central charge, and three-point forms. The general multipoint large-weight seed is an assumption supported by the five- and six-point checks recorded below, not a proved cap Ward identity. This standalone note neither inputs nor modifies the human note.

1 Conventions and scope

We use the ordinary central charge of the human note:

$$\begin{aligned} [L_m, L_n] &= (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2} - r\right)G_{m+r}, \quad \{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}, \\ c &= \frac{3}{2} + 3Q^2, \quad Q = b + b^{-1}, \quad r, s \in \mathbb{Z} + \frac{1}{2} \quad \text{in the algebra.} \end{aligned} \tag{1.1}$$

In subsequent Kac labels, r, s will instead be positive integers. We take external bottom components of intrinsically even NS primaries, and choose even highest-weight representatives for the internal modules. Upper external components and Ramond modules require separate seeds and are not included here.

Put the external weights $d_1, \dots, d_n$ at the ordered positions

$$(0, z, t_1, \dots, t_{n-4}, 1, \infty), \quad 0 < z < t_1 < \dots < t_{n-4} < 1, \tag{1.2}$$

initially on the real OPE sheet. Write $m = n - 3$ for the number of internal edges, with weights $h_1, \dots, h_m$. This follows the human note’s necklace convention: h_i are internal and d_i external. In its four-point subsection, the four external weights denoted there by h_i are consequently relabelled d_i here.

The human-note descendant and three-form conventions are

$$\begin{aligned} \mathbf{L}_{-A}\phi &= L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} \phi, & (-1)^A &= (-1)^p, \\ \rho_0(\phi_1, \phi_2, \phi_3) &= 1, & \rho_1(\phi_1, G_{-1/2}\phi_2, \phi_3) &= 1, \\ (\mathbf{G}_h)_{AB} &= \langle\langle \mathbf{L}_{-A}\phi_h \mid \mathbf{L}_{-B}\phi_h \rangle\rangle. \end{aligned} \quad (1.3)$$

The relative label on ρ_α is total descendant parity, not the absolute parity including the primaries. We keep the fixed-parity ordering of the human note; the local fusion-polynomial arguments must not be interchanged without their reflection signs.

Let $\epsilon_i \in \{0, 1\}$ select levels $\mathbb{Z}_{\geq 0} + \epsilon_i/2$ on internal edge i . The vertex labels are

$$\alpha_1 = \epsilon_1, \quad \alpha_v = \epsilon_{v-1} \oplus \epsilon_v \quad (2 \leq v \leq m), \quad \alpha_{m+1} = \epsilon_m. \quad (1.4)$$

Thus the all-bottom problem has 2^{n-3} component blocks. The ordered vertices are the left cap $\rho_{\alpha_1}(\phi_{h_1}, \phi_{d_2}, \phi_{d_1})$, the interior forms $\rho_{\alpha_v}(\phi_{h_v}, \phi_{d_{v+1}}, \phi_{h_{v-1}})$, and the right cap $\rho_{\alpha_{m+1}}(\phi_{d_n}, \phi_{d_{n-1}}, \phi_{h_m})$, with descendants inserted as appropriate. Primary structure constants are stripped off.

2 The pillow map and the sphere conformal factor

2.1 Periods, the normalized elliptic integral, and the coordinate

Use the double cover and holomorphic differential

$$y^2 = x(x-z)(x-1), \quad \omega = \frac{dx}{y}. \quad (2.1)$$

With the cycle orientations chosen to give $\text{Im } \tau > 0$, the periods and the human-note normalization of K are

$$\begin{aligned} \oint_A \omega &= 2\pi K(z), & \oint_B \omega &= 2\pi i K(1-z), \\ K(z) &= \frac{1}{\pi} \int_0^1 \frac{d\xi}{\sqrt{\xi(1-\xi)(1-z\xi)}} = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right), \end{aligned} \quad (2.2)$$

$$\tau = i \frac{K(1-z)}{K(z)}, \quad \mathbf{q} = e^{\pi i \tau}. \quad (2.3)$$

In particular, $K(0) = 1$. Set $\theta_j(\tau) = \vartheta_j(0 \mid \tau)$. When a nome is used as the argument, $\theta_3(\mathbf{q}) = \sum_{k \in \mathbb{Z}} \mathbf{q}^{k^2}$, so $\theta_3(\mathbf{q}) = \theta_3(\tau)$ and $\theta_3(\mathbf{q}^2) = \vartheta_3(0 \mid 2\tau)$. Then

$$z = \frac{\theta_2(\tau)^4}{\theta_3(\tau)^4}, \quad K(z) = \theta_3(\tau)^2, \quad w(x) = \frac{1}{\theta_3(\tau)^2} \int_0^x \frac{d\xi}{\sqrt{\xi(\xi-z)(\xi-1)}}. \quad (2.4)$$

Thus $w \sim w + 2\pi \sim w + 2\pi\tau$. On the chosen sheet the branch points map to $0, \pi, \pi + \pi\tau, \pi\tau$, respectively. The quotient $w \sim -w$ is the pillow. Write $w_j = w(t_j)$ for the mobile insertions.

2.2 Propagation factors and their branches

For $n \geq 5$, define the OPE-aligned cylinder parameters and effective propagation factors by

$$\begin{aligned} p_1 &= -e^{iw_1}, & \varrho_1 &= 4p_1, \\ p_i &= e^{i(w_i - w_{i-1})} \quad (2 \leq i \leq m-1), & \varrho_i &= p_i, \\ p_m &= -\mathbf{q}e^{-iw_{m-1}}, & \varrho_m &= 4p_m. \end{aligned} \quad (2.5)$$

They obey

$$\prod_{i=1}^m p_i = \mathfrak{q}, \quad \prod_{i=1}^m \varrho_i = 16\mathfrak{q}. \quad (2.6)$$

For $n = 4$, set $p_1 = \mathfrak{q}$ and $\varrho_1 = 16\mathfrak{q}$ instead. We use ϱ_i to distinguish propagation factors from the three-form ρ_α .

The two endpoint factors of 4 encode the cap normalization, and each interior segment has no such factor. Their product gives the same $16\mathfrak{q}$ as the four-point OPE normalization. The endpoint minus signs in (2.5) are fixed by the aligned sheet: there $w_j = \pi + iv_j$ and all p_i are positive. Take positive square roots there and continue them coherently. Since NS null levels can be half-integral, these signs cannot be discarded or reselected at recursive calls. They are geometric branch choices, not the edge-parity labels ϵ_i .

2.3 Geometric prefactor and normalized pillow block

The flat metrics are related by

$$dw d\bar{w} = e^{2\sigma} dx d\bar{x}, \quad e^{2\sigma} = \frac{1}{|\theta_3(\tau)|^4 |x(x-z)(x-1)|}. \quad (2.7)$$

A mobile bottom primary transforms as $\phi_x(t) = (dw/dt)^d \phi_w(w(t))$. Its chiral Jacobian is therefore

$$\left(\frac{dw}{dt}\right)^d = \theta_3(\tau)^{-2d} [t(t-z)(t-1)]^{-d/2}. \quad (2.8)$$

On the ordered real sheet we absorb the fixed local phase into the insertion state and write the positive radicand $t(1-t)(t-z)$ below. Combining these Jacobians with the corner factors and Weyl anomaly gives the scalar coordinate factor

$$\begin{aligned} \Lambda_n^{(c)} &= z^{c/24-d_1-d_2} (1-z)^{c/24-d_2-d_{n-1}} \\ &\times \theta_3(\tau)^{c/2-4(d_1+d_2+d_{n-1}+d_n)-2\sum_{j=1}^{n-4} d_{j+2}} \\ &\times \prod_{j=1}^{n-4} [t_j(1-t_j)(t_j-z)]^{-d_{j+2}/2}. \end{aligned} \quad (2.9)$$

This is the same bottom-primary coordinate factor as in the bosonic construction; supersymmetry changes the sewn modules, not the ordinary primary Jacobians.

Following the human note, $\mathcal{G}$ denotes the pillow cap matrix element after removing the primary propagation powers. Its sphere reconstruction is

$$\mathcal{F}_{c,\epsilon}^{\text{NS}}(\mathbf{d}, \mathbf{h}; z, \mathbf{t}) = \Lambda_n^{(c)} \prod_{i=1}^m \varrho_i^{h_i-c/24} \mathcal{G}_{n,\epsilon}(\mathbf{h}; \mathbf{p}). \quad (2.10)$$

At four points this is the parity-resolved version of the human note's definition:

$$\mathcal{G}_{4,\epsilon} = (16\mathfrak{q})^{-h+c/24} \langle \psi_{43} | \mathbb{P}_h \mathfrak{q}^{L_0-c/24} | \psi_{12} \rangle_\epsilon. \quad (2.11)$$

For the even primary term the cap overlaps multiply to $16^{h-c/24}$, as in the human note. The odd component has no level-zero term.

3 The regular part: the NS pillow product

Define

$$P(\mathbf{q}) = \prod_{j \geq 1} (1 - \mathbf{q}^{2j}). \quad (3.1)$$

In the human note's normalization, the known four-point all-bottom large-weight limits are [2]

$$\lim_{h \rightarrow \infty} \mathcal{G}_{4,0}(h; \mathbf{q}) = \frac{\theta_3(\mathbf{q}^2)}{P(\mathbf{q})^{3/4}}, \quad \lim_{h \rightarrow \infty} \mathcal{G}_{4,1}(h; \mathbf{q}) = 0. \quad (3.2)$$

The theta identity useful for checking the sphere normalization is

$$z(1-z)\theta_3(\tau)^{12} = 16\mathbf{q} P(\mathbf{q})^{12}. \quad (3.3)$$

Use the human note's common-weight variable,

$$a_1 = 0, \quad a_i = h_i - h_1 \quad (2 \leq i \leq m). \quad (3.4)$$

The multipoint proposal is

$$\boxed{\lim_{\substack{h_1 \rightarrow \infty \\ a_i \text{ fixed}}} \mathcal{G}_{n,\epsilon} = \delta_{\epsilon,0} \mathcal{C}_{\text{NS}}(\mathbf{q}), \quad \mathcal{C}_{\text{NS}}(\mathbf{q}) = \frac{\theta_3(\mathbf{q}^2)}{P(\mathbf{q})^{3/4}}.} \quad (3.5)$$

By the Jacobi triple product, this is explicitly

$$\boxed{\mathcal{C}_{\text{NS}}(\mathbf{q}) = \prod_{j=1}^{\infty} \frac{(1 - \mathbf{q}^{4j})(1 + \mathbf{q}^{4j-2})^2}{(1 - \mathbf{q}^{2j})^{3/4}}.} \quad (3.6)$$

Fractional powers are continued from $\mathbf{q} = 0$, where the product is one. Equation (3.6) replaces the bosonic pillow product $P(\mathbf{q})^{-1/2}$ while keeping the geometric factor $\Lambda_n^{(c)}$ unchanged. The dependence of the regular part on the mobile insertion positions is only through $\mathbf{q} = \prod_i p_i$. The propagation powers in (2.10) retain the individual-position dependence.

The proposed large-weight reduction is that leading normalized even interior vertices transport the same oscillator state, leaving the four-point cap contribution. Parity-changing vertices have no leading contribution in the present normalization. This motivates (3.5); a direct cap Ward-identity proof is not supplied here.

The pillow factor must not be identified with an unprojected torus trace. The ordinary NS descendant character at torus nome $\mathbf{q}^2$ is $\prod_{j \geq 1} (1 + \mathbf{q}^{2j-1})/(1 - \mathbf{q}^{2j})$. It has a term linear in $\mathbf{q}$, whereas the cap seed in (3.2) is

$$\mathcal{C}_{\text{NS}}(\mathbf{q}) = 1 + \frac{11}{4}\mathbf{q}^2 + \frac{93}{32}\mathbf{q}^4 + O(\mathbf{q}^6). \quad (3.7)$$

4 Singular part and the closed recursion

4.1 Null-vector data in the human-note convention

At fixed generic c , the NS Kac weights, levels, and null parities are

$$h_{rs}(c) = \frac{Q^2 - (rb + sb^{-1})^2}{8}, \quad \ell_{rs} = \frac{rs}{2}, \quad \pi_{rs} = rs \bmod 2, \quad r, s \geq 1, \quad r + s \in 2\mathbb{Z}. \quad (4.1)$$

The range includes $(r, s) = (1, 1)$, for which $h_{11} = 0$ and $\ell_{11} = 1/2$. The $r \geq 2$ restriction used to solve for a moving c -pole does not apply to this fixed- c h -recursion.

The inverse null-norm slope is the human note's

$$\mathbf{A}_{rs}^c = \frac{1}{2} \prod_{\substack{u=1-r, \dots, r \\ v=1-s, \dots, s \\ u+v \in 2\mathbb{Z} \\ (u,v) \neq (0,0), (r,s)}} \left[\frac{ub + vb^{-1}}{\sqrt{2}} \right]^{-1}. \quad (4.2)$$

Here b is fixed by c , not recalculated from a shifted internal weight. For any weight x , set $\lambda_x^2 = Q^2 - 8x$. Define

$$\mathbf{P}_{rs}^\alpha(x, y; c) = \prod_{\substack{u=1-r, 3-r, \dots, r-1 \\ v=1-s, 3-s, \dots, s-1 \\ u+v-r-s \in 4\mathbb{Z}+2(1-\alpha)}} \frac{\lambda_x - \lambda_y + ub + vb^{-1}}{2\sqrt{2}} \frac{\lambda_x + \lambda_y + ub + vb^{-1}}{2\sqrt{2}}, \quad \alpha \in \{0, 1\}. \quad (4.3)$$

These are precisely the $\mathbf{P}^0, \mathbf{P}^1$ of the human note. For example $\mathbf{P}_{11}^0(x, y; c) = y - x$, $\mathbf{P}_{11}^1 = 1$, and $\mathbf{A}_{11}^c = 1/2$. Their argument order matters:

$$\mathbf{P}_{rs}^0(x, y; c) = (-1)^{\lceil rs/2 \rceil} \mathbf{P}_{rs}^0(y, x; c), \quad \mathbf{P}_{rs}^1(x, y; c) = (-1)^{\lfloor rs/2 \rfloor} \mathbf{P}_{rs}^1(y, x; c). \quad (4.4)$$

For a pole on edge k , define the pole values and shifted child by

$$\begin{aligned} h_j^* &= h_j - h_k + h_{rs} = a_j - a_k + h_{rs}, \\ h_j' &= h_j^* + \ell_{rs} \delta_{jk}, \end{aligned} \quad \epsilon_j' = \epsilon_j \oplus (\pi_{rs} \delta_{jk}). \quad (4.5)$$

The two weight pairs adjacent to edge k are

$$(x_L, y_L) = \begin{cases} (d_1, d_2), & k = 1, \\ (h_{k-1}^*, d_{k+1}), & k > 1, \end{cases} \quad (x_R, y_R) = \begin{cases} (d_n, d_{n-1}), & k = m, \\ (h_{k+1}^*, d_{k+2}), & k < m. \end{cases} \quad (4.6)$$

In the oriented bottom-component convention of Section 1, the scalar residue kernel is

$$\boxed{\mathbf{R}_{rs}^{(k, \epsilon)}(\mathbf{h}^*) = (-1)^{rs} \mathbf{A}_{rs}^c \mathbf{P}_{rs}^{\alpha_k}(x_L, y_L; c) \mathbf{P}_{rs}^{\alpha_{k+1}}(x_R, y_R; c).} \quad (4.7)$$

The sign is tied to the human note's fixed-parity three-form ordering, not an extra universal sign to append to an independently normalized formula. An odd null vector flips the edge label and both adjacent vertex labels; an even null vector flips neither. This ordered kernel is the one used in the numerical check described below.

4.2 Regular part plus recursively factorized residues

The common-weight pole is $h_1 = h_{rs} - a_k$. At the pole,

$$\text{Res}_{h_1=h_{rs}-a_k} \mathcal{G}_{n, \epsilon} = \varrho_k^{\ell_{rs}} \mathbf{R}_{rs}^{(k, \epsilon)}(\mathbf{h}^*) \mathcal{G}_{n, \epsilon'}(\mathbf{h}'). \quad (4.8)$$

The propagation factor follows by comparing the primary powers before and after replacing the null state by its primary of weight $h_{rs} + \ell_{rs}$:

$$\frac{\prod_j \varrho_j^{h_j^* + \ell_{rs} \delta_{jk} - c/24}}{\prod_j \varrho_j^{h_j^* - c/24}} = \varrho_k^{\ell_{rs}}. \quad (4.9)$$

The geometric factor is independent of internal weights and cancels. The null-state matrix elements can therefore be evaluated in the plane with the human-note fusion polynomials and transformed back.

At a fixed elliptic order, assume coefficientwise rationality in h_1 and generic, noncoincident simple poles. Subtracting their principal parts leaves a polynomial in h_1 . The finite large- h_1 limit in (3.5) fixes that polynomial to the constant regular part. Consequently the proposed recursion is

$$\boxed{\mathcal{G}_{n,\epsilon}(\mathbf{h}) = \underbrace{\delta_{\epsilon,0}\mathcal{C}_{\text{NS}}(\mathbf{q})}_{\text{regular part in } h_1} + \sum_{k=1}^m \sum_{\substack{r,s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{\varrho_k^{rs/2} \mathbf{R}_{rs}^{(k,\epsilon)}(\mathbf{h}^*)}{h_k - h_{rs}(c)} \mathcal{G}_{n,\epsilon'}(\mathbf{h}').} \quad (4.10)$$

The sum is the singular part. There is no separate $o(1)$ remainder: finite-weight corrections are supplied by the pole terms. For the all-even component the regular part is the infinite product, and for the other components it is zero. “Regular” refers to internal-weight dependence, not to an expansion order in $\mathbf{q}$.

4.3 The unit-seed elliptic block and reconstruction

As in the human note, reserve H for the block with the regular product removed:

$$H_{n,\epsilon} := \mathcal{C}_{\text{NS}}(\mathbf{q})^{-1} \mathcal{G}_{n,\epsilon}. \quad (4.11)$$

It obeys

$$\boxed{H_{n,\epsilon}(\mathbf{h}) = \delta_{\epsilon,0} + \sum_{k=1}^m \sum_{\substack{r,s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{\varrho_k^{rs/2} \mathbf{R}_{rs}^{(k,\epsilon)}(\mathbf{h}^*)}{h_k - h_{rs}(c)} H_{n,\epsilon'}(\mathbf{h}').} \quad (4.12)$$

All external weights, c , and coordinates remain fixed. The internal differences for the next call are obtained from the child tuple $\mathbf{h}'$; they are not frozen to their parent values. Since $\mathcal{C}_{\text{NS}}$ is weight independent, there is no additional ratio of theta functions in the kernel. The final sphere block is

$$\boxed{\mathcal{F}_{c,\epsilon}^{\text{NS}} = \Lambda_n^{(c)} \prod_i \varrho_i^{h_i - c/24} \mathcal{C}_{\text{NS}}(\mathbf{q}) H_{n,\epsilon}.} \quad (4.13)$$

5 Four- and five-point specializations

For four points, $\varrho_1 = 16\mathbf{q}$ and $\alpha_1 = \alpha_2 = \epsilon$. Thus

$$H_{4,\epsilon}(h) = \delta_{\epsilon,0} + \sum_{\substack{r,s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{(16\mathbf{q})^{rs/2} (-1)^{rs} \mathbf{A}_{rs}^c \mathbf{P}_{rs}^\epsilon(d_1, d_2; c) \mathbf{P}_{rs}^\epsilon(d_4, d_3; c)}{h - h_{rs}} H_{4,\epsilon \oplus \pi_{rs}}(h_{rs} + rs/2). \quad (5.1)$$

The odd component is understood in the human-note phase convention. As a useful sign check, the $(1,1)$ term gives $H_{4,1} = -2\mathbf{q}^{1/2}/h + \dots$; in the plane this corresponds to the leading coefficient $-1/(2h)$ multiplying $z^{h-d_1-d_2+1/2}$ in the same convention.

For five points, the weights at $(0, z, t, 1, \infty)$ are $(d_1, d_2, d_3, d_4, d_5)$ and

$$\varrho_1 = -4e^{iw(t)}, \quad \varrho_2 = -4qe^{-iw(t)}, \quad a = h_2 - h_1. \quad (5.2)$$

For each (r, s) define

$$\begin{aligned} \mathbf{R}_{rs}^{(1)} &= (-1)^{rs} \mathbf{A}_{rs}^c \mathbf{P}_{rs}^{\epsilon_1}(d_1, d_2; c) \mathbf{P}_{rs}^{\epsilon_1 \oplus \epsilon_2}(h_{rs} + a, d_3; c), \\ \mathbf{R}_{rs}^{(2)} &= (-1)^{rs} \mathbf{A}_{rs}^c \mathbf{P}_{rs}^{\epsilon_1 \oplus \epsilon_2}(h_{rs} - a, d_3; c) \mathbf{P}_{rs}^{\epsilon_2}(d_5, d_4; c). \end{aligned} \quad (5.3)$$

The complete two-edge recursion is

$$\begin{aligned} H_{5, \epsilon_1 \epsilon_2}(h_1, h_2) &= \delta_{\epsilon_1, 0} \delta_{\epsilon_2, 0} \\ &+ \sum_{\substack{r, s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{\varrho_1^{rs/2} \mathbf{R}_{rs}^{(1)}}{h_1 - h_{rs}} H_{5, \epsilon_1 \oplus \pi_{rs}, \epsilon_2}(h_{rs} + rs/2, h_{rs} + a) \\ &+ \sum_{\substack{r, s \geq 1 \\ r+s \in 2\mathbb{Z}}} \frac{\varrho_2^{rs/2} \mathbf{R}_{rs}^{(2)}}{h_2 - h_{rs}} H_{5, \epsilon_1, \epsilon_2 \oplus \pi_{rs}}(h_{rs} - a, h_{rs} + rs/2). \end{aligned} \quad (5.4)$$

Its full sphere prefactor is (2.9) with $n = 5$, and $\mathcal{C}_{\text{NS}}$ is unchanged. No new function of t is inserted into the regular part.

6 Implementation, checks, and status

To evaluate a component block, compute $(\mathbf{q}, w_j, p_i, \varrho_i)$ on a coherent sheet, form the labels (1.4), and apply (4.12) using (4.2)–(4.7). Each recursive term lowers the requested level on edge k by $rs/2$, so a finite-order coefficient calculation terminates. A twice-level integer index avoids floating-point comparisons of half-integers. Finally multiply by (4.13). Coincident Kac poles or coincident common-weight pole locations require limits or a separate confluent treatment.

6.1 Independent PBW audit: definition of the comparison

The direct calculation uses only the NS algebra (1.1), the primary Ward identities, and the three-form normalization (1.3). It does not call either c - or h -recursion, or use Kac weights or fusion polynomials. At each level ℓ_i it constructs the PBW Gram matrix G_i and the ordered three-point matrices. The plane coefficient is their contraction,

$$B_\ell = \beta_R^\top G_m^{-1} V_{m-1} G_{m-1}^{-1} \cdots V_1 G_1^{-1} \beta_L. \quad (6.1)$$

Here V_i has bra weight h_{i+1} , inserted weight d_{i+2} , and ket weight h_i ; β_L, β_R are the cap vectors in the ordering specified in Section 1. For one edge this reduces to $\beta_R^\top G_1^{-1} \beta_L$. All parity sectors are retained, with $\epsilon_i = 2\ell_i \bmod 2$.

Set $x_1 = z/t_1$, $x_i = t_{i-1}/t_i$ for $2 \leq i < m$, and $x_m = t_{m-1}$; for four points use $x_1 = z$. Then the plane expansion is

$$\mathcal{F}_{c, \epsilon}^{\text{NS}} = z^{h_1 - d_1 - d_2} \prod_{j=1}^{m-1} t_j^{h_{j+1} - h_j - d_{j+2}} \sum_{\ell_i \in \mathbb{Z}_{\geq 0 + \epsilon_i/2}} B_\ell \prod_i x_i^{\ell_i}. \quad (6.2)$$

We substitute the pillow map and divide by the entire prefactor in (4.13); thus the quantity tested is precisely H , not a block in a different normalization.

For reproducibility, the coordinate expansion is generated to the requested order from products, rather than from a fixed low-order polynomial. Write $U_j = \prod_{i \leq j} p_i$, $V_j = \prod_{i > j} p_i$, so $U_j V_j = \mathbf{q}$, and define

$$Z := \frac{z}{16\mathbf{q}} = \prod_{k \geq 1} \frac{(1 + \mathbf{q}^{2k})^8}{(1 + \mathbf{q}^{2k-1})^8}, \quad (6.3)$$

$$Y_j := \frac{t_j}{4V_j} = (1 + U_j)^2 \prod_{k \geq 1} \frac{(1 + \mathbf{q}^{2k})^4}{(1 + \mathbf{q}^{2k-1})^4} \frac{(1 + U_j \mathbf{q}^{2k})^2 (1 + V_j \mathbf{q}^{2k-1})^2}{(1 + U_j \mathbf{q}^{2k-1})^2 (1 + V_j \mathbf{q}^{2k-2})^2}. \quad (6.4)$$

The coordinate units are $x_1/(4p_1) = Z/Y_1$, $x_i/p_i = Y_{i-1}/Y_i$ in the interior, and $x_m/(4p_m) = Y_{m-1}$. Their constant terms are one. The multiplier applied to the composed plane descendant series, after the primary powers cancel, is

$$\begin{aligned} & Z^{h_1 - c/24} \prod_{j=1}^{m-1} Y_j^{h_{j+1} - h_j} \prod_{j=1}^{m-1} [(1 - t_j)(1 - z/t_j)]^{d_{j+2}/2} \\ & \times (1 - z)^{-c/24 + d_2 + d_{n-1}} \theta_3(\mathbf{q})^{-c/2 + 4(d_1 + d_2 + d_{n-1} + d_n) + 2 \sum_j d_{j+2}} \mathcal{C}_{\text{NS}}(\mathbf{q})^{-1}. \end{aligned} \quad (6.5)$$

Consequently the check tests the coordinate map, all conformal factors, the fermionic signs, the regular product, and the residue recursion together. No effective shift of c is used in this implementation.

“Through degree N ” means every coefficient of $\prod_i p_i^{\ell_i}$ with $\ell_i \geq 0$ half-integral and $\sum_i \ell_i \leq N$. It is not a rectangular cutoff on each edge. The number of tested coefficients is $\binom{2N+m}{m}$, including the constant and the vanishing coefficients. At $N = 10$ the largest individual Gram matrix has dimension 161.

As an independent algebra check, the new PBW/Ward implementation was compared with the existing exact descendant-three-form implementation through total level three: all 57 Gram entries and 110 Ward entries agree identically for independent symbols c, h, d, k .

6.2 Analytic PBW comparison through total degree three

All parameters are kept symbolic, with coefficients in $\mathbb{Q}(b, d_1, \dots, d_n, h_1, \dots, h_m)$. The PBW calculation first uses an independent c , and only at the comparison stage substitutes $c = 3/2 + 3(b + b^{-1})^2$. Thus an exact comparison here does not mean evaluation at a special rational parameter set.

There are two equivalent symbolic tests. One expands both sides of (4.12) and reduces every coefficient difference to zero. The other gives a smaller rational-identity certificate. Write $C_\ell = [\prod_i p_i^{\ell_i}]H$ and $\kappa_i = \varrho_i/p_i$. For every coefficient computed by PBW, check:

1. Its reduced denominator, viewed over $\mathbb{Q}(b)$, divides

$$\prod_{k=1}^m \prod_{\substack{r, s \geq 1, \\ r+s \in 2\mathbb{Z} \\ rs \leq 2\ell_k}} (h_k - h_{rs}). \quad (6.6)$$

Thus no additional poles or higher-order poles are left unchecked.

2. Every residue equals the proposed kernel times the lower-degree PBW coefficient:

$$\operatorname{Res}_{h_k=h_{rs}} C_\ell = \kappa_k^{rs/2} \mathbf{R}_{rs}^{(k,\epsilon)} \Big|_{h_k=h_{rs}} C_{\ell-(rs/2)\mathbf{e}_k} \Big|_{h_k=h_{rs}+rs/2}. \quad (6.7)$$

In this local test the other h_j are independent symbolic variables held fixed. Substitution of their pole values h_j^* then gives exactly the common-weight residue in (4.12).

3. The directly computed regular part is

$$\lim_{L \rightarrow \infty} C_\ell(h_1 + L, \dots, h_m + L) = \delta_{\ell, \mathbf{0}}. \quad (6.8)$$

This is tested from numerator and denominator degrees in L , including their leading coefficients; the seed is not supplied to the PBW side.

With the differences $h_j - h_1$ fixed, the difference from the recursive answer is a rational function with no poles and zero regular part. It therefore vanishes identically. Induction in total degree makes these three checks an exact coefficient comparison with the expanded recursion, not merely a check of its pole locations.

The completed generic-symbolic audit gives

Points	Coefficients	Exact method	Result
4	7	Expanded coefficient differences	All zero
5	28	Expanded differences and certificate	All zero
6	84	Pole and regular-part certificate	All zero

The five-point certificate checks all 106 candidate residue identities, and the six-point certificate all 342, including vanishing residues. Every coefficient also passes the denominator-divisibility and large-common-weight regular-part tests. There is no numerical tolerance or specialization of b, d_i, h_i in these results. No change to the proposed recursion or its conformal factor is required by this audit.

For example, both independent calculations give

$$\begin{aligned} H_{5,00} &= 1 - \frac{2(d_1 - d_2)(d_3 + h_1 - h_2)}{h_1} p_1 + \frac{2(d_4 - d_5)(d_3 - h_1 + h_2)}{h_2} p_2 + O_{\text{tot}}(2), \\ H_{5,10} &= -\frac{\sqrt{p_1}}{h_1} + O_{\text{tot}}(3/2), \quad H_{5,01} = -\frac{\sqrt{p_2}}{h_2} + O_{\text{tot}}(3/2), \\ H_{5,11} &= \frac{d_3 - h_1 - h_2}{h_1 h_2} \sqrt{p_1 p_2} + O_{\text{tot}}(2). \end{aligned} \quad (6.9)$$

Here $O_{\text{tot}}(a)$ contains terms of total degree at least a . These signs refer throughout to the human-note three-form convention. The six-point leading terms additionally distinguish an interior edge from a cap edge:

$$\begin{aligned} H_{6,010} &= -\frac{\sqrt{p_2}}{2h_2} + O_{\text{tot}}(3/2), \\ H_{6,101} &= \frac{\sqrt{p_1 p_3}}{h_1 h_3} + O_{\text{tot}}(2), \\ H_{6,111} &= -\frac{(d_3 - h_1 - h_2)(d_4 - h_2 - h_3)}{2h_1 h_2 h_3} \sqrt{p_1 p_2 p_3} + O_{\text{tot}}(5/2). \end{aligned} \quad (6.10)$$

In particular, the interior edge uses $\varrho_2 = p_2$, not $4p_2$.

6.3 Numerical PBW comparison through total degree ten

Four generic parameter sets were tested for both five and six points. The six-point fixtures are listed below; the five-point fixture deletes the fourth external entry and the third internal entry. In this table $c = 3/2 + 3(b + b^{-1})^2$, and the labels A–D refer to the PBW audit, not the earlier degree-four fixtures below. The new degree-ten c -recursion audit uses these same A–D fixtures.

Case	b	$(d_1, d_2, d_3, d_4, d_5, d_6)$	(h_1, h_2, h_3)
A	1.27	(.31, .42, .53, .37, .47, .28)	(.73, 1.10, 1.37)
B	1.43	(.22, .61, .39, .58, .74, .45)	(1.13, .85, 1.69)
C	.83	(.17, .83, 1.21, .46, .34, .67)	(.19, 1.47, .82)
D	1.61	(1.12, .26, .79, 1.33, .58, .91)	(2.31, .64, 1.09)

For each set, every coefficient through total degree ten was compared: 231 coefficients in all four parity sectors for five points, and 1771 coefficients in all eight sectors for six points. Using 80-digit arithmetic, the maximum scaled errors are

PBW case	Five points	Six points
A	4.84×10^{-74}	1.80×10^{-69}
B	9.34×10^{-72}	7.81×10^{-69}
C	1.94×10^{-71}	7.57×10^{-69}
D	3.37×10^{-70}	2.42×10^{-67}

Let u_ℓ and v_ℓ denote the coefficient of the same H computed by PBW and recursion, respectively. The error is

$$\frac{|u_\ell - v_\ell|}{\max\{1, |u_\ell|, |v_\ell|\}}. \quad (6.11)$$

Every coefficient passes the 10^{-50} tolerance. Individual edge levels can reach ten; the comparison does not restrict each edge to the evenly shared cutoff. The general numerical audit includes the four-point control at case A: all 21 coefficients through degree ten agree with maximum scaled error 7.01×10^{-70} .

Repeating six-point case C at 110-digit precision gives maximum scaled error 1.48×10^{-97} for the same 1771 coefficients. Comparing the 80- and 110-digit runs directly, the PBW coefficients change by at most 3.04×10^{-74} , and the recursion coefficients by at most 7.57×10^{-69} in the same scaled norm. This precision improvement is consistent with finite-precision cancellation rather than a residual disagreement in the formula.

These tests include the regular product, whose first nonconstant term occurs at total p_i -degree $2m$: four for five points and six for six points. Thus the degree-ten audit is stronger than a low-order comparison that never sees the proposed NS regular product.

6.4 Reproducing the PBW audit

The implementation lives in `Code/sphere_five_kummer_h_recursion/`. The independent sewing engine is `ns_pillow_direct_pbw.py`; the coordinate products and proposed recursion are in `ns_pillow_elliptic_audit.py`. From that directory run

```
python3 check_ns_pillow_direct_pbw.py
--mode symbolic --points 5 --degree 3

python3 check_ns_pillow_direct_pbw.py
--mode symbolic --symbolic-method residues --points 6 --degree 3

python3 check_ns_pillow_direct_pbw.py
--mode numeric --points 6 --degree 10 --case A --dps 80
```

Each indented line continues the command above it. Repeat the numerical command for both five and six points and cases A–D. Use `--case C --dps 110` for the precision repeat and `--mode oracle` for the independent low-level Gram/Ward audit. Appending `--output result.json` saves the coefficient ledger. The saved audit files are under `Data Set/h-Recursion/`, with names beginning `ns_pillow_pbw_`. The numerical files contain both coefficient values, precision, error by degree and parity, and the minimum visited recursion denominator. The symbolic files retain an exact-zero status for each coefficient and, for the certificate method, the individual pole and regular-part checks.

The five quick regression tests also check signs, truncation consistency, and rejection of deliberately incorrect residues, regular parts, and extra poles. The numerical PBW fixtures have positive-definite Gram matrices and use diagonally scaled Cholesky solves. This audit does not claim to implement a general nonunitary or complex-weight Gram solver.

6.5 Independent literature c -recursion: conventions and seed

The independent comparison uses the sphere linear-channel recursion of Belavin–Geiko [3], equations (3.12)–(3.18), with the global matrix elements of their Appendix A. The four-point control uses their equations (3.8)–(3.10). There is an indexing caveat in arXiv v1: the interior line of (3.18) prints the adjacent external labels as d_{i+2}, d_{i+3} . With the h_i ordering in their Figure 3 and (3.15), the adjacent insertions are instead d_{i+1}, d_{i+2} , with the corresponding component labels. We use this graph-consistent adjacency, not the literal shifted indices of that line. For six points, for example, the middle edge joins insertions d_3, d_4 . This indexing correction is separate from a normalization change.

Their central charge, denoted c_{BG} here only to make the translation explicit, satisfies

$$c = \frac{3}{2}c_{\text{BG}}, \quad c_{rs}(h) = \frac{3}{2}c_{rs}^{\text{BG}}(h), \quad J_{rs}(h) := -\frac{dc_{rs}(h)}{dh} = \frac{3}{2}J_{rs}^{\text{BG}}(h). \quad (6.12)$$

All subsequent formulas use only the human-note c . Converting a pole but not its Jacobian would change the residue by a factor of $3/2$. The three-form reflection phases and odd-null transport are likewise translated into (1.3); no second normalization of H is introduced.

The paper’s parameters q_i in its equation (3.16) are plane plumbing ratios, not $\mathbf{q}$ or the pillow p_i . After reversing its external and internal ordering to (1.2), they are the reverse-ordered list of our x_i in (6.2). For example, at six points their list becomes $(t_2, t_1/t_2, z/t_1) = (x_3, x_2, x_1)$. Their normalized plane series is therefore compared with H only after restoring the primary powers and dividing by the entire prefactor in (4.13), as in the PBW pullback above.

For clarity, the implemented coefficient recursion in our ordering is

$$B_\ell^\epsilon(c, \mathbf{h}) = B_\ell^{\text{glob}, \epsilon}(\mathbf{h}) + \sum_{k=1}^m \sum_{\substack{r \geq 2, s \geq 1 \\ r+s \in 2\mathbb{Z} \\ rs \leq 2\ell_k}} \frac{J_{rs}(h_k) \mathcal{R}_{rs}^{(k, \epsilon)}(\mathbf{h}; c_*)}{c - c_*} B_{\ell - (rs/2)\mathbf{e}_k}^{\epsilon'} \left(c_*, \mathbf{h} + \frac{rs}{2} \mathbf{e}_k \right), \quad c_* = c_{rs}(h_k). \quad (6.13)$$

Here $\mathbf{e}_k$ is the k th unit vector, $\epsilon'_j = \epsilon_j \oplus (\pi_{rs} \delta_{jk})$, and B_ℓ^ϵ denotes the coefficient of the plane descendant series with its leading powers removed. With the adjacent pairs of (4.6), now using the unshifted neighboring h_j rather than h_j^* , the residue is

$$\mathcal{R}_{rs}^{(k, \epsilon)}(\mathbf{h}; c_*) = (-1)^{rs} \mathbf{A}_{rs}^{c_*} \mathbf{P}_{rs}^{\alpha_k}(x_L, y_L; c_*) \mathbf{P}_{rs}^{\alpha_{k+1}}(x_R, y_R; c_*). \quad (6.14)$$

Every null-vector factor is evaluated at the moving central-charge pole. One convenient branch, appropriate to the generic fixtures below, is

$$X_{rs}(h) := b_{rs}(h)^2 = -\frac{4h + rs - 1 + \sqrt{16h^2 + 8(rs-1)h + (r-s)^2}}{r^2 - 1}, \quad c_{rs}(h) = \frac{15}{2} + 3(X_{rs}(h) + X_{rs}(h)^{-1}). \quad (6.15)$$

The range $r \geq 2$ is essential for this moving- c parametrization. Only h_k shifts in its child, and c changes to c_* . In contrast, (4.5) shifts the entire internal-weight tuple at fixed c . Thus this is not a rearrangement of the implemented elliptic recursion.

The large- c seed is a genuine global $\mathfrak{osp}(1|2)$ network. Writing $\ell_i = N_i + \epsilon_i/2$ and $v_i = L_{-1}^{N_i} G_{-1/2}^{\epsilon_i} \phi_{h_i}$, it is

$$B_\ell^{\text{glob}, \epsilon} = \frac{\rho_{\alpha_1}(v_1, \phi_{d_2}, \phi_{d_1}) \prod_{i=1}^{m-1} \rho_{\alpha_{i+1}}(v_{i+1}, \phi_{d_{i+2}}, v_i) \rho_{\alpha_{m+1}}(\phi_{d_n}, \phi_{d_{n-1}}, v_m)}{\prod_{i=1}^m N_i! (2h_i)_{N_i + \epsilon_i}}. \quad (6.16)$$

The matrix elements are evaluated algebraically in the global algebra, with precisely the human-note ordering of ρ_α . This seed is neither the constant 1 nor $\mathcal{C}_{\text{NS}}$. In particular, the c -recursive plane calculation does not assume the proposed large-internal-weight pillow seed.

The independent code is `Code/c_Recursion/ns_multipoint_c_recursion.py`, using `ns_recursion_recipe.py` for moving poles and scalar residues and `ns_global_osp_block.py` for the global seed. None calls the elliptic recursion or a PBW solver. The new comparison driver only adds memoization and selects the vertex labels for each edge-parity tuple. The exact pillow map and geometric multiplier are shared with the preceding PBW audit; the two plane-block constructions are not.

6.6 Numerical literature comparison through total degree ten

The new test recomputes both the plane c -recursion and the elliptic h -recursion at 80-digit precision, using all four fixtures in Section 6.3. It compares their coefficients of the same unit-seed H , with the scaled error (6.11). The cutoff is $\sum_i \ell_i \leq 10$, including every half-integer edge level: 231 coefficients

per five-point fixture in four parity sectors, and 1771 per six-point fixture in eight sectors. The maximum errors are

Case	Five points: c versus h	Six points: c versus h
A	4.14×10^{-74}	1.80×10^{-69}
B	9.34×10^{-72}	7.81×10^{-69}
C	1.94×10^{-71}	7.57×10^{-69}
D	3.37×10^{-70}	2.42×10^{-67}

Every coefficient passes the 10^{-50} tolerance; no modification of the proposed h -recursion or its prefactor is required. The four-point control at case A also passes all 21 coefficients, with maximum error 1.41×10^{-69} .

As a third check, the newly computed c -recursive H is compared with the saved independent PBW coefficient ledgers. The maximum errors across A–D are 1.47×10^{-73} for five points and 3.58×10^{-74} for six points. These saved PBW values are used only as comparison targets, never to construct either recursion. Both recursions are computed afresh at each precision.

Repeating six-point case C at 110 digits gives maximum error 1.48×10^{-97} between the two recursions, and 2.34×10^{-104} between c -recursion and the independent PBW ledger. Across the 80- and 110-digit runs, the transformed c -recursive coefficients change by at most 1.28×10^{-74} and the h -recursive coefficients by 7.57×10^{-69} in the same scaled norm. The improvement is consistent with finite-precision cancellation, not an order-ten discrepancy in the proposal.

The comparison includes the nonconstant terms of the NS product: $\mathfrak{q}^2 = (\prod_i p_i)^2$ first appears at total degree four for five points and six for six points. Since the independent c -recursive seed is (6.16), the agreement checks the proposed pillow regular product, rather than assuming it on both sides.

From Code/sphere_five_kummer_h_recursion/, run

```
python3 check_ns_pillow_c_recursion.py
--points 6 --case A --degree 10 --dps 80
--output result.json
```

Repeat with five points and cases A–D; case C at 110 digits provides the precision repeat. The optional `--pbw-ledger` argument takes the corresponding saved PBW JSON path and enables the third comparison. It is not needed for c versus h . The new files in **Data Set/h-Recursion/** begin with `ns_pillow_c_comparison_`. They retain every plane c -recursive coefficient, its transformed H , the independently computed h -recursive H , per-degree and per-parity errors, precision, and source hashes. The summary JSON collects all runs and the precision comparison. These are our numerical evaluations of the published recursion, not numerical data supplied by the paper’s authors.

Five additional regression tests check that memoization leaves the unmodified literature implementation unchanged, exercise all six-point sectors at degree three, verify the nonunit global seed and odd phase, reject invalid level tuples, and detect an intentionally omitted Jacobian conversion. The 13 existing multipoint c -recursion regression tests also pass. The numerical test assumes generic nonconfluent moving c -poles; it does not prescribe a limiting procedure at collisions.

6.7 Earlier comparison with the NS c -recursion

The existing reproducible five-point test compares the elliptic proposal against the independent multipoint NS c -recursion [3], pulled back by the exact pillow coordinate series. It checks all four parity channels through total elliptic degree four: 45 coefficients per parameter set, including half-integer edge powers. This is not a rectangular level-four check and not a direct PBW computation. The generic cases are

Case	c	$(d_1, d_2, d_3, d_4, d_5)$	(h_1, h_2)
A	14.7	(0.31, 0.42, 0.53, 0.47, 0.28)	(0.73, 1.10)
B	21.3	(0.22, 0.61, 0.39, 0.74, 0.45)	(1.13, 0.85)

At 75-digit working precision, with 60-digit comparison data, the maximum relative errors were

Case	Maximum relative error for H
A	2.7×10^{-60}
B	1.9×10^{-60}

The error denominator is $\max(1, |\text{left}|, |\text{right}|)$. A separate large-weight check used $(h_1, h_1 + 0.37)$ in case A and $(h_1, h_1 - 0.28)$ in case B, with $h_1 = 50, 200, 800$. The coefficientwise deviations from the proposed seed decreased approximately as h_1^{-1} . For example, the all-even $p_1^2 p_2^2$ coefficient of H in case A was $-0.0971729251, -0.0253007025, -0.0063915533$, approaching 0.

The product identity in (3.6) was separately checked symbolically through $\mathfrak{q}^{10}$. The prefactor conversion at $\mathfrak{q} = 0.01, 0.1, 0.3$ agreed to better than 2.3×10^{-75} . Those separate tests by themselves check normalization; the actual degree-ten block-coefficient comparisons are in Sections 6.3 and 6.6. From `Code/sphere_five_kummer_h_recursion/`, rerun the checks with:

```
python3 check_ns_sphere_five_elliptic_h_recursion.py
```

The general- n seed remains the proposal in (3.5). The direct PBW and independent c -recursion checks support the four-, five-, and six-point formulas at the stated finite orders; they do not prove the seed for arbitrary n or control the remainder at a fixed nonzero coordinate. The bosonic multipoint elliptic construction is prior work of Artemev and Khromov [1], in particular their equations (3.16) and (5.14)–(5.18). The four-point NS seed is prior work of Hadasz, Jaskólski, and Suchanek [2], Section 6. We do not claim either as new, nor make a novelty claim for this multipoint NS proposal without a separate literature audit.

References

- [1] A. Artemev and D. Khromov, *WKB-asymptotics for multipoint Virasoro conformal blocks and applications*, arXiv:2603.08194v2.
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- [3] V. A. Belavin and R. V. Geiko, *c -Recursion for multi-point superconformal blocks. NS sector*, arXiv:1806.09563.